

EDUCATION

- **Korea Advanced Institute of Science and Technology** Daejeon, South Korea
Doctor of Philosophy in Civil and Environmental Engineering
Dissertation: Deep-Learning based Urban Vehicle Trajectory Analytics
Mar. 2017 – Aug. 2021
Advisor: Hwasoo Yeo
- **Korea Advanced Institute of Science and Technology** Daejeon, South Korea
Master of Science in Civil and Environmental Engineering
Thesis: Development of Simulation-based Lane Change Control for Autonomous Vehicles
Sep. 2015 – Feb. 2017
Advisor: Hwasoo Yeo
- **Korea Advanced Institute of Science and Technology** Daejeon, South Korea
Bachelor of Science in Civil and Environmental Engineering
Feb. 2011 – Aug. 2015

CAREER

- **Assistant Professor** University of Minnesota
Department of Civil, Environmental, and Geo- Engineering
◦ **Research Focus:**
 - * AI/GenAI for transportation and mobility systems
 - * Spatiotemporal data modeling
 - * Connected Automated Vehicles & Cooperative-ITSJan. 2024 – present
- **Teaching:**
 - * CEGE 8490 - Generative AI in Transportation Research (graduate) - 2025 Fall
 - * CEGE 4160/5180 - Applied Machine Learning for Civil, Environmental, and Geo-Engineers - 2025 Spring (undergraduate/graduate)
 - * CEGE 3101 - Computer Applications I (undergraduate) - 2024 Spring

SELECTED PUBLICATIONS

- [S1] Ryu, Seunghee, Kwon, Donghoon, et al. (2025). *BO4Mob: Bayesian Optimization Benchmarks for High-Dimensional Urban Mobility Problem*. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, December 2025. Corresponding author
- [S2] **Choi, Seongjin**, Jin, Zhixiong, et al. (2025). “A gentle introduction and tutorial on deep generative models in transportation research”. In: *Transportation Research Part C: Emerging Technologies* 176, p. 105145. <https://doi.org/10.1016/j.trc.2025.105145>
- [S3] **Choi, Seongjin**, Saunier, Nicolas, Zheng, Vincent Zhihao, et al. (2025). “Scalable Dynamic Mixture Model with Full Covariance for Probabilistic Traffic Forecasting”. In: *Transportation Science* 59.4. Available at arXiv: <https://arxiv.org/abs/2212.06653>, pp. 708–720. <https://doi.org/10.1287/trsc.2024.0547>
- [S4] **Choi, Seongjin**, Lee, Jinwoo (2023a). “Optimal planning of parking infrastructure and fleet size for Shared Autonomous Vehicles”. In: *Transportation Research Part E: Logistics and Transportation Review* 176, p. 103213. <https://doi.org/10.1016/j.trc.2023.103213>
- [S5] **Choi, Seongjin**, Lee, Donghoun, et al. (2023). “Framework for Connected and Automated Bus Rapid Transit with Sectionalized Speed Guidance based on deep reinforcement learning: Field test in Sejong City”. In: *Transportation Research Part C: Emerging Technologies* 148, p. 104049. <https://doi.org/10.1016/j.trc.2023.104049> [**Choi & Lee: equal contribution**]
- [S6] **Choi, Seongjin**, Kim, Jiwon, Yeo, Hwasoo (2021). “TrajGAIL: Generating urban vehicle trajectories using generative adversarial imitation learning”. In: *Transportation Research Part C: Emerging Technologies* 128. Source code available at <https://github.com/benchoi93/TrajGAIL>, p. 103091. <https://doi.org/10.1016/j.trc.2021.103091>

PREVIOUS CAREER

- **Postdoctoral Researcher at Smart Transportation Lab**

Fully funded from McGill University

Supervised by Prof. Lijun Sun

McGill University

- **Project:** Bridging Data-Driven and Behavioral Models for Transportation (IVADO)
Responsibility: Spatiotemporal Forecasting, Spatiotemporal Residual Correction, Multi-agent reinforcement learning for cooperative adaptive cruise control (CACC), Deep Generative Models
Collaborators: Prof. Nicolas Saunier (Polytechnique Montreal), Professor Martin Trepanier (Polytechnique Montreal), Professor Francesco Ciari (Polytechnique Montreal)

Jan. 2022 – Dec. 2023

- **Postdoctoral Researcher at AIxMobility Lab**

Fully funded from KAIST

Supervised by Prof. Hwasoo Yeo

KAIST

- **Project:** Development of Multi-level Traffic Simulation for C-ITS and CAV
Responsibility: Designing overall system architecture and data structure, Developing core functions including lane-changing-model and car-following-model, Deploying cloud-based simulator using Microsoft Azure, Developing scenario generator based on real-time data collected in Sejong city
Collaborators: Dr. Sehyun Tak (Korea Transport Institute), Dr. Donghoun Lee (Korea Transport Institute)
- **Project:** Development of Cloud-based Demand-Responsive Routing System
Responsibility: Designing cloud system architecture on Microsoft Azure, Optimizing computation for real-time applications

Sep. 2021 - Nov. 2021

- **Visiting scholar at the University of Queensland**

Funded from BK21+ (Brain Korea 21 Plus)

Supervised by Prof. Jiwon Kim

University of Queensland

- **Project:** Deep learning applications in urban vehicle trajectory analytics
Responsibility: Research conceptualization, Problem design, model development, result analysis, writing and editing
Achievements: 2 journal papers, 2 conference presentations

May. 2017 – Aug. 2017

Sep. 2018 – Nov. 2018

FUNDED PROJECTS (TOTAL FUNDING: \$1,317,104; \$699,624 FOR CHOI)

- University of Minnesota Center for Transportation Studies “Embedding Human-Understandable Symbolic Rules into Vision-Language Decision Systems for Safe, Transparent, and Ethical Autonomous Driving,” \$75,000, (02/01/2026-01/31/2027) **PI Choi**
- Minnesota Department of Transportation (MnDOT) “Img2Speed: Generative AI and Multimodal Machine Learning for Predicting Operating Speed Distributions from Roadway Design and Context,” \$166,846, (08/01/2026-06/30/2028) **PI Choi**
- University of Minnesota, Data Science Initiative (DSI), GenAI4Science Seed Projects. “Physics-informed Generative Model for High-Resolution Traffic State Estimation and Optimal Sensor Placement,” \$67,110 (01/01/2026-12/31/2026) **PI Choi**
- Minnesota Department of Transportation (MnDOT) “Detector Data Artificial Intelligence (AI),” \$84,712, (10/01/2025-09/30/2026) **PI Choi**
- Korea Transport Institute “Research on Search-Augmented Generation Techniques Based on Traffic Simulation for Numerical AI,” \$21,045 (30M KRW), (05/01/2025-02/28/2026) **PI Choi**
- National Research Foundation of Korea “K-health-mobility: Multi-layer digital twin of healthcare and mobility and supplementary system design for vulnerable groups using a synthetic population of an aging society,” \$670,391 (960M KRW) (\$240,275 for Choi), (03/01/2025-03/01/2029), PI Woo, **Co-PI Choi**

- Minnesota Department of Transportation (MnDOT) and Local Road Research Board (LRRB) “Estimating Likely Mode Shift and VMT Reduction Potential using TBI Data and AI Algorithms,” \$198,000 (\$10,636 for Choi), (07/01/2025 - 06/30/2027), PI Khani **Co-PI Choi**
- University of Minnesota Center for Transportation Studies, “CTS Infrastructure Grant”, \$9,000, **PI Choi**
- University of Minnesota Center for Transportation Studies, “Digitizing Minnesota Transportation Observatory Data”, \$10,000, (07/01/2024 – 06/30/2026), **PI Choi**
- University of Minnesota Center for Transportation Studies Faculty Fellows, “General Framework for Modeling Probabilistic Traffic Data Imputation and Forecasting as Inverse Problem using Deep Generative Prior”, \$15,000, (07/01/2024 – 06/30/2026), **PI Choi**

PROFESSIONAL SERVICES

- **Early Career Editorial Board Member**
Transportation Research Part C: Emerging Technologies 2026-present
- **Member**
CTS Transportation Safety and Mobility Council 2024-present
- **Member**
Transportation Research Board
AED50(7) Subcommittee on Edge Computing 2025
- **Member**
Transportation Research Board
AED50(6) Subcommittee on Artificial Intelligence Ethics and Equity 2024-2025
- **Editorial Member**
Journal of Korean Society of Transport 2023
- **Program Committee**
Socially Interactive Autonomous Mobility (SIAM) 2024
- **Guest Editor**
Journal of Advanced Transportation (Special Issue)
Advanced Data Intelligence Theory and Practice in Transportation 2023 2023
- **Reviewer**
Transportation Research Part C: Emerging Technologies
IEEE Transactions on Intelligent Transportation Systems
IEEE Open Journal of Intelligent Transportation Systems
Scientific Data
Transportation Research Records
Energy
EURO Journal on Transportation and Logistics
Physica A: Statistical Mechanics and its Applications
Sensors
IEEE International Conference on Intelligent Transportation Systems
IEEE Intelligent Vehicles Symposium
Transportation Research Board Annual Meeting

ADVISING EXPERIENCE

- Guoliang Feng, Postdoctoral Researcher, Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Christopher Cheong, Ph.D. student, Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Lindong Liu, Ph.D. student, Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.

- Pouya Parsa, Ph.D. student (from Fall 2026; completed M.S.), Department of Civil, Environmental, and Geo-Engineering, University of Minnesota.
- Ross Volkov, Master student, Data Science Program, University of Minnesota.
- Mitch Kosieradzki, Master student, Data Science Program, University of Minnesota.
- Saravanan Rammesh Adhav, Master student, Robotics, University of Minnesota.
- Christopher Herr, Master student, Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Amirhossein Kiani, Ph.D. student (Ph.D. Oral Preliminary Exam Committee), Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Xiaohuan Zeng, Ph.D. student (Ph.D. Oral Preliminary Exam/Defense Committee), Department of Geography, University of Minnesota.
- Arian Zare, Ph.D. student (Ph.D. Oral Preliminary Exam Committee), Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Behnam Davazdah Emami, Ph.D. student (Ph.D. Defense Committee), Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Kwangho Baek, Ph.D. student (Ph.D. Oral Preliminary Exam/Defense Committee), Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.
- Tianyi Li, Ph.D. student (Ph.D. Defense Committee), Department of Civil, Environmental, and Geo- Engineering, University of Minnesota.

FULL LIST OF PUBLICATIONS

— PUBLICATIONS UNDER REVIEW

- Choi, Seongjin**, Yoo, Sunbin, Lee, Sugie (2026). “Do E-Scooter Speed Governance Policies Reduce Harsh Acceleration and Deceleration? Evidence from 19.5 Million Trips Around a Regulatory Ban”. Available at arXiv: <https://arxiv.org/abs/2604.26236>.
- Choi, Seongjin** (2026). “Beyond Coauthorship: Semantic Structure and Phantom Collaborators in Transportation Research, 1967–2025”. Available at arXiv: <https://arxiv.org/abs/2604.23699>.
- Min, Donggyu, **Choi, Seongjin**, Kim, Dong-Kyu (2026a). “Deep Reinforcement Learning for Dynamic Origin-Destination Matrix Estimation in Microscopic Traffic Simulations Considering Credit Assignment”. Submitted to Transportation Research Part C.
- Ryu, Seunghee, Kang, Seungmo, **Choi, Seongjin** (2026a). “Multi-Fidelity Surrogate-Based Optimization for Passenger-Centric Bus Berth Allocation in Urban Transit Hubs”. Submitted to Transportation Research Part C.

— INTERNATIONAL JOURNAL

- Choi, Seongjin**, Jin, Zhixiong, Ham, Seung Woo, Kim, Jiwon, Sun, Lijun (2025). “A gentle introduction and tutorial on deep generative models in transportation research”. In: *Transportation Research Part C: Emerging Technologies* 176, p. 105145. <https://doi.org/10.1016/j.trc.2025.105145>.
- Choi, Seongjin**, Saunier, Nicolas, Zheng, Vincent Zhihao, Trépanier, Martin, Sun, Lijun (2025). “Scalable Dynamic Mixture Model with Full Covariance for Probabilistic Traffic Forecasting”. In: *Transportation Science* 59.4. Available at arXiv: <https://arxiv.org/abs/2212.06653>, pp. 708–720. <https://doi.org/10.1287/trsc.2024.0547>.
- Choi, Seongjin**, Lee, Jinwoo (2023a). “Optimal planning of parking infrastructure and fleet size for Shared Autonomous Vehicles”. In: *Transportation Research Part E: Logistics and Transportation Review* 176, p. 103213. <https://doi.org/10.1016/j.tre.2023.103213>.

- Choi, Seongjin**, Lee, Donghoun, Kim, Sari, Tak, Sehyun (2023). “Framework for Connected and Automated Bus Rapid Transit with Sectionalized Speed Guidance based on deep reinforcement learning: Field test in Sejong City”. In: *Transportation Research Part C: Emerging Technologies* 148, p. 104049. <https://doi.org/10.1016/j.trc.2023.104049>.
- Choi, Seongjin**, Kim, Jiwon, Yeo, Hwasoo (2021). “TrajGAIL: Generating urban vehicle trajectories using generative adversarial imitation learning”. In: *Transportation Research Part C: Emerging Technologies* 128. Source code available at <https://github.com/benchoi93/TrajGAIL>, p. 103091. <https://doi.org/10.1016/j.trc.2021.103091>.
- Kosieradzki, Mitch, **Choi, Seongjin** (2026a). “TrajFlow: A Generative Framework for Occupancy Density Estimation Using Normalizing Flows”. In: *Artificial Intelligence for Transportation*. In press. Available at arXiv: <https://arxiv.org/abs/2501.14266>.
- Cheong, Christopher, Davis, Gary, **Choi, Seongjin** (2026). “Weaver: Kronecker Product Approximations of Spatiotemporal Attention for Traffic Network Forecasting”. In: *Transportation Research Part C: Emerging Technologies*. In press. Available at arXiv: <https://arxiv.org/abs/2511.08888>.
- Liu, Lindong, Jin, Zhixiong, **Choi, Seongjin** (2026). “PMA-Diffusion: A Physics-guided Mask-aware Diffusion Framework for Traffic State Estimation from Sparse Observations”. In: *Transportation Research Part C: Emerging Technologies* 190. Available at arXiv: <https://arxiv.org/abs/2512.06183>, p. 105801. <https://doi.org/10.1016/j.trc.2026.105801>.
- Yoo, Sunbin, **Choi, Seongjin**, Kim, Dohyung (2026). “Can Renewable Energy Meet the Surging Power Demand of Artificial Intelligence? A Systematic Review”. In: *Renewable and Sustainable Energy Reviews* 240, p. 117213. <https://doi.org/10.1016/j.rser.2026.117213>.
- Cho, Jungwoo, **Choi, Seongjin** (2026). “Toward Safe Integration of UAM in Terminal Airspace: UAM Route Feasibility Assessment Using Probabilistic Aircraft Trajectory Prediction”. In: *IEEE Transactions on Intelligent Transportation Systems*. Early Access, pp. 1–10. <https://doi.org/10.1109/TITS.2026.3651399>.
- Kim, Yeeun, **Choi, Seongjin**, Jeon, Sujae, Yeo, Hwasoo (2026). “NextSim: Multi-Level Traffic Simulation for Urban Networks Using Dynamic Level Assignment”. In: *IEEE Transactions on Intelligent Transportation Systems*. Early Access, pp. 1–15. <https://doi.org/10.1109/TITS.2026.3671220>.
- Lin, Tengfeng, Jin, Zhixiong, **Choi, Seongjin**, Yeo, Hwasoo (2025). “Predicted lane-based time to collision: A novel surrogate safety measure for pedestrian potential risk evaluation at non-signalized intersections”. In: *Transportation Research Interdisciplinary Perspectives* 34, p. 101705. <https://doi.org/10.1016/j.trip.2025.101705>.
- Zheng, Vincent Zhihao, **Choi, Seongjin**, Sun, Lijun (2025). “Probabilistic Traffic Forecasting with Dynamic Regression”. In: *Transportation Science* 59.4. Available at arXiv: <https://arxiv.org/abs/2301.06650>, pp. 689–707. <https://doi.org/10.1287/trsc.2024.0560>.
- Zheng, Vincent Zhihao, **Choi, Seongjin**, Sun, Lijun (2024a). “Better batch for deep probabilistic time series forecasting”. In: *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*. Vol. 238. Proceedings of Machine Learning Research. <https://proceedings.mlr.press/v238/zheng24a.html>, pp. 91–99.
- Jin, Zhixiong, Kim, Jiwon, Yeo, Hwasoo, **Choi, Seongjin** (2022). “Transformer-based map-matching model with limited labeled data using transfer-learning approach”. In: *Transportation Research Part C: Emerging Technologies* 140. Corresponding author, p. 103668. <https://doi.org/10.1016/j.trc.2022.103668>.
- Tak, Sehyun, **Choi, Seongjin** (2022). “Safety Monitoring System of CAVs Considering the Trade-Off between Sampling Interval and Data Reliability”. In: *Sensors* 22.10. Corresponding author, p. 3611. <https://doi.org/10.3390/s22103611>.
- Kim, Yeeun, **Choi, Seongjin**, Yeo, Hwasoo (2020). “Extended Urban Cell Transmission Model Using Agent-based Modeling”. In: *Procedia Computer Science* 170, pp. 354–361. <https://doi.org/10.1016/j.procs.2020.03.058>.

- Choi, Seongjin**, Kim, Jiwon, Yeo, Hwasoo (2019a). “Attention-based Recurrent Neural Network for Urban Vehicle Trajectory Prediction”. In: *Procedia Computer Science* 151, pp. 327–334. <https://doi.org/10.1016/j.procs.2019.04.046>.
- Lee, Donghoun, Tak, Sehyun, **Choi, Seongjin**, Yeo, Hwasoo (2019a). “Development of Risk Predictive Collision Avoidance System and Its Impact on Traffic and Vehicular Safety”. In: *Transportation Research Record: Journal of the Transportation Research Board* 2673.7, pp. 454–465. <https://doi.org/10.1177/0361198119836972>.
- Kim, Yeeun, **Choi, Seongjin**, Park, Jihyuk, Yeo, Hwasoo (2019a). “Agent-based Mesoscopic Urban Traffic Simulation based on Multi-lane Cell Transmission Model”. In: *Procedia Computer Science* 151, pp. 240–247. <https://doi.org/10.1016/j.procs.2019.04.035>.
- Choi, Seongjin**, Yeo, Hwasoo, Kim, Jiwon (2018a). “Network-Wide Vehicle Trajectory Prediction in Urban Traffic Networks using Deep Learning”. In: *Transportation Research Record: Journal of the Transportation Research Board* 2672.45, pp. 173–184. <https://doi.org/10.1177/0361198118794735>.
- Choi, Seongjin**, Suh, Jonghae, Yeo, Hwasoo (2017). “Microscopic Analysis of Climbing Lane Performance at Freeway Uphill Section”. In: *Transportation Research Procedia* 21, pp. 98–109. <https://doi.org/10.1016/j.trpro.2017.03.081>.

— PEER-REVIEWED INTERNATIONAL CONFERENCE

- Ryu, Seunghee, Kwon, Donghoon, **Choi, Seongjin**, Deshwal, Aryan, Kang, Seungmo, Osorio, Carolina (2025). *BO4Mob: Bayesian Optimization Benchmarks for High-Dimensional Urban Mobility Problem*. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, December 2025. Corresponding author.
- Liu, Lindong, **Choi, Seongjin** (2026). *PMA-diffusion: A Physics-guided Mask-aware Diffusion Framework for Traffic State Estimation from Sparse Observations*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Kosieradzki, Mitch, **Choi, Seongjin** (2026b). *TrajFlow: A Generative Framework for Occupancy Density Estimation Using Normalizing Flows*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Kwon, Donghoon, Kang, Seungmo, **Choi, Seongjin** (2026). *TrafficNetQA: Question Answering Benchmark for Evaluating LLM Performance on Traffic Network Files*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Ryu, Seunghee, Kang, Seungmo, **Choi, Seongjin** (2026b). *Multi-Fidelity Surrogate-Based Optimization for Passenger-Centric Bus Berth Allocation in Urban Transit Hubs*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Choi, Seongjin**, Jin, Zhixiong, Ham, Seung Woo, Kim, Jiwon, Sun, Lijun (2026). *A Gentle Introduction to Deep Generative Models in Transportation Research*. The 105th Transportation Research Board Annual Meeting, January 2026.
- Ryu, Seunghee, Kwon, Donghoon, **Choi, Seongjin**, Deshwal, Aryan, Kang, Seungmo, Osorio, Carolina (2026). *BO4Mob: Bayesian Optimization Benchmarks for High-Dimensional Urban Mobility Problem*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Min, Donggyu, **Choi, Seongjin**, Kim, Dong-Kyu (2026b). *Deep Reinforcement Learning for Dynamic Origin-Destination Matrix Estimation in Microscopic Traffic Simulations Considering Credit Assignment*. The 105th Transportation Research Board Annual Meeting, January 2026.
- Parsa, Pouya, Li, Keya, Kockelman, Kara M., **Choi, Seongjin** (2026). *Video-based Vehicle Surveillance in the Wild: License Plate, Make, and Model Recognition with Self Reflective Vision-Language Models*. The 105th Transportation Research Board Annual Meeting, January 2026. Corresponding author.
- Liu, Lindong, **Choi, Seongjin** (2025). *PMA-diffusion: A Physics-guided Mask-aware Diffusion Framework for Traffic State Estimation from Sparse Observations*. NeurIPS Workshop on Urban AI, December 2025. Corresponding author.

- Jiang, Sicong, Huang, Zilin, Qian, Kangan, Luo, Ziang, Zhu, Tianze, Zhong, Yang, Tang, Yihong, Kong, Menglin, Wang, Yunlong, Jiao, Siwen, Ye, Hao, Sheng, Zihao, Zhao, Xin, Wen, Tuopu, Fu, Zheng, Chen, Sikai, Jiang, Kun, Yang, Diange, **Choi, Seongjin**, Sun, Lijun (2025). *A Survey on Vision-Language-Action Models for Autonomous Driving*. ICCV 2025 Workshop on Distillation of Foundation Models for Autonomous Driving, October 2025. Available at <https://arxiv.org/abs/2506.24044>. GitHub: <https://github.com/JohnsonJiang1996/Awesome-VLA4AD>.
- Kwon, Donghoon, Kang, Seungmo, **Choi, Seongjin** (2025). *TrafficNetQA: Question Answering Datasets for Evaluating LLM Performance on Traffic Network Files*. International Conference on Advances in Geographic Information Systems 2025 (ACM SIGSPATIAL 2025), November 2025. Corresponding author.
- Parsa, Pouya, Li, Keya, Kockelman, Kara M., **Choi, Seongjin** (2025). *Video-based Vehicle Surveillance in the Wild: License Plate, Make, and Model Recognition with Self Reflective Vision-Language Models*. Bridging Transportation Researchers Conference 7, August 2025. Corresponding author.
- Liu, Fuqiang, Jiang, Sicong, Miranda-Moreno, Luis, **Choi, Seongjin**, Sun, Lijun (2025). *Adversarial Vulnerabilities in Large Language Models for Time Series Forecasting*. AISTATS 2025, May 2025. Corresponding author.
- Cheong, Christopher, **Choi, Seongjin** (2025). *Weaver: A Spatio-Temporal Deep Learning Model Architecture Based on the Mixed Kronecker Matrix-Vector Identity*. The 104th Transportation Research Board Annual Meeting, January 2025. Corresponding author.
- Liu, Fuqiang, Jiang, Sicong, Miranda-Moreno, Luis, **Choi, Seongjin**, Sun, Lijun (2024). *Adversarial Vulnerabilities in Large Language Models for Time Series Forecasting*. NeurIPS 2024 Safe Generative AI Workshop, December 2024. Corresponding author.
- Choi, Seongjin**, Rodriguez, Sergio, Osorio, Carolina (2024). *Bayesian Optimization for High-dimensional Urban Mobility Problems*. NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty, December 2024.
- Li, Tianyi, Wang, Shian, Shang, Mingfeng, **Choi, Seongjin**, Stern, Raphael (2024). *A Customizable Neural Network based Framework for Autonomous Vehicle Control with Human-Guided Learning*. IEEE Intelligent Transportation Systems Conference (ITSC) 2024 Workshop on Large-Scale Smart Mobility, September 2024.
- Zheng, Vincent Zhihao, **Choi, Seongjin**, Sun, Lijun (2024b). *Better batch for deep probabilistic time series forecasting*. The 27th International Conference on Artificial Intelligence and Statistics (AISTATS), May 2024.
- Choi, Seongjin**, Lee, Jinwoo (2024). *An Analytical Model for Parking Infrastructure and Fleet Size Planning in Shared Autonomous Vehicle Systems with Spatio-Temporal Heterogeneity*. The 103rd Transportation Research Board Annual Meeting, January 2024.
- Jiang, Sicong, **Choi, Seongjin**, Sun, Lijun (2024). *Communication-Aware Reinforcement Learning for Cooperative Adaptive Cruise Control*. The 103rd Transportation Research Board Annual Meeting, January 2024.
- Zheng, Vincent Zhihao, **Choi, Seongjin**, Sun, Lijun (2024c). *Deep Probabilistic Traffic Forecasting with Correlated Errors*. The 103rd Transportation Research Board Annual Meeting, January 2024.
- Kim, Yeeun, **Choi, Seongjin**, Jeon, Sujae, Yeo, Hwasoo (2024). *Multi-Level Traffic Simulation Using Dynamic Simulation Level Assignment for Urban Network*. The 103rd Transportation Research Board Annual Meeting, January 2024.
- Choi, Seongjin**, Lee, Jinwoo (2023b). *Analytical Parking Planning Model with Shared Autonomous Vehicles*. The 102nd Transportation Research Board Annual Meeting, January 2023.
- Lin, Tengfeng, **Choi, Seongjin**, Jin, Zhixiong, Yeo, Hwasoo (2023). *Evaluation of Pedestrian's Potential Risk at Non-signalized Intersection Based on Predicted Post-Encroachment Time using Deep Learning Methods*. The 102nd Transportation Research Board Annual Meeting, January 2023.
- Choi, Seongjin**, Saunier, Nicolas, Trepanier, Martin, Sun, Lijun (2022). *Spatiotemporal Residual Regularization with Kronecker Product Structure for Traffic Forecasting*. Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS 2022), Workshop on Gaussian Processes, Spatiotemporal Modeling, and Decision-making Systems, December 2022.

- Lin, Tengfeng, Jin, Zhixiong, **Choi, Seongjin**, Yeo, Hwasoo (2022). *A Framework for Pedestrian Sub-classification and Arrival Time Prediction at Signalized Intersection Using Preprocessed Lidar Data*. The 101st Transportation Research Board Annual Meeting, January 2022.
- Jin, Zhixiong, **Choi, Seongjin**, Yeo, Hwasoo (2022). *Transformer-based Map Matching with Model Limited Ground-Truth Data using Transfer-Learning Approach*. TRB 2022 Annual Meeting, January 2022.
- Choi, Seongjin**, Kim, Jiwon, Park, Min Ju, Yeo, Hwasoo (2021). *TrajGAIL: Generating Urban Trajectories using Generative Adversarial Imitation Learning*. The 100th Transportation Research Board Annual Meeting, January 2021.
- Choi, Seongjin**, Yeo, Hwasoo, Kim, Jiwon (2020). *Incorporating Network Traffic State for Urban Vehicle Trajectory Prediction*. The 99th Transportation Research Board Annual Meeting, January 2020.
- Choi, Seongjin**, Kim, Jiwon, Yu, Hwapyeong, Yeo, Hwasoo (2019). *Real-time Prediction of Arterial Vehicle Trajectories: An Application to Predictive Route Guidance for an Emergency Vehicle*. 2019 IEEE Intelligent Transportation Systems Conference (ITSC), October 2019.
- Kim, Yeeun, **Choi, Seongjin**, Park, Jihyuk, Yeo, Hwasoo (2019b). *Agent-based Mesoscopic Urban Traffic Simulation based on Multi-lane Cell Transmission Model*. The 10th International Conference on Ambient Systems, Networks and Technologies, Acadia University, May 2019.
- Choi, Seongjin**, Kim, Jiwon, Yeo, Hwasoo (2019b). *Attention-based Recurrent Neural Network for Urban Vehicle Trajectory Prediction*. The 10th International Conference on Ambient Systems, Networks and Technologies, Acadia University, May 2019.
- Lee, Donghoun, Tak, Sehyun, **Choi, Seongjin**, Yeo, Hwasoo (2019b). *Development of risk predictive collision avoidance system and its impact on traffic and vehicular safety*. The 98th Transportation Research Board Annual Meeting, January 2019.
- Kim, Yeeun, **Choi, Seongjin**, Yeo, Hwasoo (2018). *Incorporation of Driver Distraction in Car-following model based on Driver's Eye Glance Behavior*. 2018 21st International Conference on Intelligent Transportation Systems (ITSC), October 2018.
- Choi, Seongjin**, Yeo, Hwasoo, Kim, Jiwon (2018b). *Network-wide Vehicle Trajectory Prediction in Urban Traffic Networks Using Deep Learning*. The 97th Transportation Research Board Annual Meeting, January 2018.
- Choi, Seongjin**, Tak, Sehyun, Kim, Jihu, Yeo, Hwasoo (2017). *Traffic Event Classification using Convolutional Neural Network*. The 30th KKHTCNN Symposium on Civil Engineering, November 2017.
- Tak, Sehyun, Yeo, Hwasoo, Kim, Yeeun, **Choi, Seongjin** (2017). *A Study on the Dynamics of Driver Vision Transitions and its Impacts on Vehicle Safety*. 10th SHRP 2 Safety Data-Symposium: From Analysis to Results, October 2017.
- Tak, Sehyun, Lee, Donghoun, **Choi, Seongjin**, Yeo, Hwasoo (2017). *Collision Avoidance System with Uni-directional Communication for Mitigating the Adverse Effects on Following Vehicles*. Urban Transport 2017, September 2017.
- Choi, Seongjin**, Yeo, Hwasoo (2017). *Framework for simulation-based lane change control for autonomous vehicles*. IEEE Intelligent Vehicles Symposium (IV), June 2017.
- Tak, Sehyun, **Choi, Seongjin**, Lee, Donghoun, Yeo, Hwasoo (2017). *A Comparison Analysis of Track-Based Train Operation System and Communication-Based Train Operation System for Train Safety*. The 96th Transportation Research Board Annual Meeting, January 2017.
- Tak, Sehyun, **Choi, Seongjin**, Yeo, Hwasoo (2016). *The Effect of Communication and GPS Uncertainty on Safety Performance of Communication-based Train Control*. The 1st Asian Conference on Railway Infrastructure and Transportation, October 2016.
- Choi, Seongjin**, Suh, Jonghae, Yeo, Hwasoo (2016). *Microscopic Analysis of Climbing Lane Performance at Freeway Uphill Section*. 2016 International Symposium of Transport Simulation, June 2016.

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Choi, Seongjin, Kim, Jiwon, Yu, Hwapyeong, Ka, Dongho, Yeo, Hwasoo (2019). "Deep-learning based urban vehicle trajectory prediction". In: *Journal of Korean Society of Transportation* 37.5, pp. 422–429.

- Kim, Yeeun, **Choi, Seongjin**, Yeo, Hwasoo (2019). “A study on development of a car-following model for accident simulation caused by driver distraction”. In: *Journal of Korean Society of Transportation* 37.1, pp. 39–50.
- Suh, Jonghae, **Choi, Seongjin**, Yeo, Hwasoo (2018). “A Study on Climbing Lane in Freeway Uphill Segment by Developing a Microscopic Traffic Simulation Model”. In: *Journal of Korean Society of Transportation* 36.4, pp. 263–273.

INVITED TALKS

- **UC Berkeley PATH seminar**
Berkeley (virtual) 2025.11.18
- **Transportation Research in the Era of Generative AI** NCITE/CTS Symposium
Minneapolis, MN 2025.10.10
- **Deep Generative Models for Transportation and Mobility Data** Virginia Tech
Virtual 2025.03.24
- **I-24 MOTION workshop**
A Gentle Introduction and Tutorial on Deep Generative Models in Transportation Research 2025.01.08
Washington, D.C.
- **2024 Future-Shaping ACE Congress**
Traffic Forecasting with Correlated Spatiotemporal Structure Korea University
Virtual 2024.10.28
- **Leveraging Spatiotemporally Correlated Structure for Deep-Learning-based Traffic Forecasting** Kookmin University
Seoul, South Korea 2024.06.28
- **Deep Generative Models for Transportation and Mobility Data** Seoul National University
Seoul, South Korea 2024.06.18
- **The 2nd Cho Chun Shik Graduate School of Mobility Summer Camp**
Deep Generative Models for Transportation and Mobility Data — Intro & Tutorial KAIST
Jeju, South Korea 2024.06.21
- **Warren Distinguished Lecture**
Understanding Distributions of Traffic and Mobility Data University of Minnesota
Minneapolis, MN, USA 2024.02.23
- **NSF-IRES**
Leveraging Spatiotemporally Correlated Structure for Deep-Learning-based Traffic Forecasting Texas A&M
Virtual 2024.02.01
- **Deep Generative Models Tutorial** KAIST
Virtual 2023.11.02
- **Deep Generative Models for Mobility Trajectory Data** Western University
London, Canada 2023.08.02
- **Learning the Distribution of Traffic and Mobility Data** KAIST
Daejeon, South Korea 2023.05.31
- **Learning the Distribution of Traffic and Mobility Data** Seoul National University
Seoul, South Korea 2023.05.23
- **2022 Future-Shaping ACE Congress**
Deep Generative Models for Transportation and Mobility Data Korea University
Virtual 2022.11.21
- **Trending Research Topics in AI and Potential Applications in Transportation Domain: (Part 2) Representation Learning** Korea Transport Institute
Virtual 2022.11.01
- **Trending Research Topics in AI and Potential Applications**

- **Modeling Behaviors of Road Users by Imitation Learning**
Seoul, South Korea

Korean Society of Road Engineers
2021.11.19

AWARDS & SCHOLARSHIPS

- **TRANSFOR22 Data Competition - 3rd place** Award
TRB AED50 Artificial Intelligence and Advanced Computing Applications 2022
- **BK21 PLUS (Brain Korea 21 PLUS) Long-term Overseas Training Program (University of Queensland)** Scholarship
Korea Advanced Institute of Science and Technology 2017
- **National Scholarship (Fully-Funded Doctoral Course)** Scholarship
Korea Advanced Institute of Science and Technology 2017-2021
- **Railroad Specialized Graduate School Scholarship** Scholarship
Ministry of Land, Infrastructure, and Transport in Republic of Korea 2015-2017
- **National Scholarship (Fully-Funded Master Course)** Scholarship
Korea Advanced Institute of Science and Technology 2015-2017
- **National Scholarship (Fully-Funded Bachelor Course)** Scholarship
Korea Advanced Institute of Science and Technology 2011-2015

PATENTS (DOMESTIC IN KOREA)

- **10-22795860000: Server and Method for Managing Shared Autonomous Vehicles**
A management method for shared autonomous vehicles. This method proposes a demand-predictive management system for shared autonomous vehicles. It covers from the prediction algorithm and server framework to handle real-time data to predict the demand hot-spots to the route planning algorithm based on Artificial Intelligence.
- **10-2018-0141857: Traffic Simulator for Evaluation of ITS systems**
A hybrid traffic simulator that can simulate the urban traffic networks in both mesoscopic and microscopic scales. This simulator uses distributed computing to simulate multiple scenarios in small amount of time, and aims to be applied to many applications such as route optimization of shared autonomous vehicles and signal control based on reinforcement learning.

PROGRAMMING SKILLS

- **Programming Languages:** Python, R, C++, Julia
- **Deep learning:** Pytorch, Tensorflow
- **Other skills:** Microsoft Azure, Docker, QGIS